

TactTail: Expanding Multimodal, Nonverbal Communication Between Human and Dog

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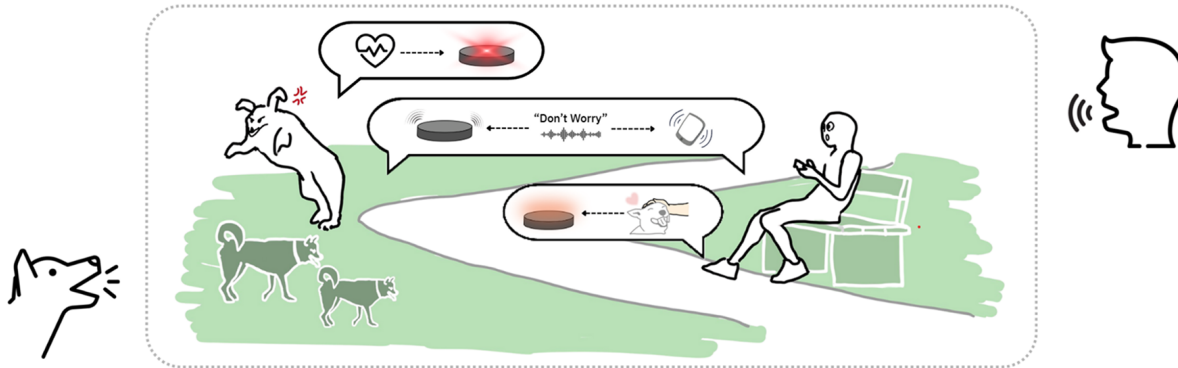


Figure 1: Illustration of TactTail’s bidirectional communication loop: the dog’s elevated arousal (e.g., increased heart rate) is detected by the wearable and signaled to the owner through subtle tactile feedback on their phone. The owner responds via app, which is translated into calming haptic and thermal feedback delivered back to the dog, supporting mutual understanding beyond verbal commands. This system enables resilient interaction even across physical distance or in noisy environments where spoken communication is ineffective.

Abstract

We present *TactTail*, a wearable multimodal system enabling bidirectional, embodied communication between humans and dogs. TactTail integrates vibrotactile, thermal, and visual feedback with real-time physiological sensing, allowing owners to deliver non-verbal cues and perceive the dog’s emotional state through subtle tactile signals on their phone. By combining speech-to-haptic translation, tactile illusions, and multimodal training, the system establishes a shared tactile language that complements or replaces verbal commands, fostering resilient interaction and deeper human-dog connection.

CCS Concepts

• Human-centered computing → Haptic devices.

Keywords

Animal-Computer Interaction, Haptics, Tactile Communication, Multimodal Interaction, Wearable Devices

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UIST Adjunct '25, Busan, Republic of Korea

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ACM ISBN 979-8-4007-2036-9/25/09

<https://doi.org/10.1145/3746058.3758976>

ACM Reference Format:

Rachel Kim and Eunho Kim. 2025. TactTail: Expanding Multimodal, Non-verbal Communication Between Human and Dog. In *The 38th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '25)*, September 28–October 01, 2025, Busan, Republic of Korea. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3746058.3758976>

1 Introduction

Communication between humans relies on shared language for complex auditory and visual exchange, whereas interaction with companion animals like dogs occurs without a shared linguistic system and is often reduced to simple auditory commands like “sit” or “stay.” This reliance on sound assumes auditory channels are always available — yet in many real-world situations, they are not. Environmental noise [7], physical distance [4], or conditions where the owner cannot speak all limit the effectiveness of voice commands. Yet dogs perceive their environment through multiple senses, including touch, temperature, smell, and visual cues, offering an opportunity to expand human-dog communication beyond language-based, human-centric channels.

Prior work in animal-computer interaction (ACI) has explored pet wearables for activity tracking or simple vibration cues, while human-computer interaction (HCI) research shows that embodied, non-verbal feedback (e.g., haptic, thermal signals) can effectively convey information or emotion, especially when language is insufficient [9]. However, most existing systems treat such feedback

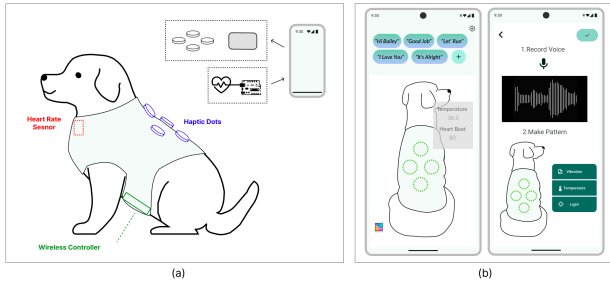


Figure 2: System overview: (a) TactTail prototype showing the placement of haptic dots, heart rate sensor, and wireless controller. (b) User interface allowing selection or authoring of corresponding spatiotemporal haptic patterns combined with audio input.

as supplementary to verbal commands [2, 4, 5, 7], rather than exploring it as an independent, primary channel for cross-species interaction.

In this work, we introduce TactTail, a wearable multimodal vest enabling bidirectional, embodied communication between humans and dogs. TactTail integrates vibrotactile, thermal, and visual feedback, allowing owners to deliver non-verbal cues such as commands or reassurance, while also sensing the dog’s arousal state. Rather than substituting voice, TactTail reframes haptic feedback as a core interaction modality, opening new avenues for richer, more resilient human-animal communication.

2 System Overview

TactTail is a haptic-based wearable system designed to enhance bidirectional communication between dogs and their owners in both remote and proximity settings. The TactTail system consists of two main components: the Dog Suit and the Owner Client Application.

2.1 Dog Suit Components

For the pet suit system (Figure 2a), we leverage the DataFeel DevKit, which integrates vibrotactile, visual, and thermal modules to deliver multisensory feedback. To extend physiological sensing, we incorporate a photoplethysmography (PPG)-based heart rate sensor connected via an Arduino microcontroller. The El Jefe controller manages power and communication, interfacing wirelessly with a mobile app that sends user commands, which are translated into vibration patterns, light signals, or thermal outputs on the dog’s body. Together, these components enable real-time bidirectional communication.

2.2 Owner Client Application

The mobile-based owner client application acts as the central interface for bidirectional communication with the pet suit via Bluetooth.

Speech to Tactile Language: Users can convert speech into spatiotemporal haptic feedback through the system’s mobile interface (Figure 2b). The system analyzes the acoustic features of the owner’s speech, and converts its wave characteristics into temporal haptic patterns, drawing on prior work in vocal-to-haptic

mapping [3]. To generate spatial patterns, the system combines sequential activations across the four linear tactors using tactile illusions [1, 8], creating a perceivable sense of directional movement. Users can also incorporate thermal feedback, authoring combined vibration and temperature patterns that leverage thermal illusions to enrich the tactile language. Users can trigger this tactile language alongside their speech. The patterns play simultaneously on the suit hardware (spatiotemporal vibration+thermal) and the user’s phone (temporal vibration), establishing a shared tactile channel between human and dog, directly inherited from the speech signal.

Real-time Status Notification and Visualization: The system continuously monitors the dog’s physiological and behavioral state, including heart rate and motion, and visualizes this information on the mobile interface in real time. When notable changes are detected—such as signs of stress, discomfort, or unusual inactivity—the system delivers context-specific tactile signals to the owner’s phone. These signals encourage the owner to notice, interpret, and gradually attune to their dog’s needs and states, fostering a more nuanced and responsive form of communication over time.

3 Use Case Scenario

1. Setting up the shared language concept: TactTail enables owners and dogs to develop a shared tactile language that complements or even replaces verbal communication in challenging environments. Building on multimodal training [2, 7], dogs learn to associate haptic and thermal patterns with familiar commands, following safety protocols to ensure comfort [6]. Once established, this shared vocabulary enables richer, bidirectional interaction.

2. Bidirectional communication loop: At the core of TactTail is a continuous communication loop. The vest monitors the dog’s physiological and behavioral state, translating changes into subtle signals such as LED color shifts and gentle vibrations delivered to the owner’s phone. These cues allow the owner to perceive the dog’s emotional or physical state (e.g., stress, restlessness) even when apart. In turn, the owner can respond by sending tactile or thermal cues, or by speaking through the app, which converts speech into spatiotemporal haptic patterns the dog can understand.

3. Example Scenarios: For example, when a dog left alone at home signals discomfort, the owner receives a quiet vibration notification and can send back a reassuring warmth pattern through the vest. Or, in a noisy outdoor setting, the owner can say “sit” into the app, and the system translates this into a directional vibration pattern the dog has learned to recognize.

4. Fostering Attunement and Connection :By supporting this ongoing, embodied exchange, TactTail fosters not just command delivery, but mutual attunement and connection—offering a resilient, expressive communication channel that deepens the human-dog bond beyond the limits of verbal interaction.

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